

# Natural Language Processing Applications

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# Session 1

These slides are prepared by the instructor, with grateful acknowledgement of James Allen and many others who made their course materials freely available online.

# Session Content

- Objective of course
- What will we learn in this course?
- Text books and Reference books
- Evaluation Plan
- Application areas of Natural Language Processing

# Objective of course

| | |
|---|---|
| CO1 | Identify and describe algorithms for real life NLP Applications |
| CO2 | Demonstrate understanding of algorithms by using different NLP tools |
| CO3 | Apply NLP techniques in state of art applications like Machine Translation, Information Extraction including Named entity recognition and Relation extraction |
| CO4 | Evaluate different approaches of implementing NLP applications along with ethical considerations |

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# What you will learn in this course

- Spellcheckers and Grammar checkers
- Question Answering and Conversational AI
- Knowledge Graph Applications
  - Including RAG variants
- Machine Translation
- Information Extraction
- Sentiment Analysis and Opinion Mining

**All applications will be discussed using implementation of Hybrid approaches of NLP(Statistical, Gen AI, Agentic AI etc.)**

**Privacy & Ethics in NLP, Scalable & Efficient NLP Systems will also be discussed**

| | |
|---|---|
| LO1 | To provide students with the knowledge on designing and applying algorithms for real life NLP Applications |
| LO2 | To give an overview of the major technologies used in NLP and hands-on experience of using such tools |
| LO3 | To apply NLP techniques in state of art applications like Machine Translation, Information Extraction including Named entity recognition and Relation extraction |
| LO4 | To give students opportunities to sharpen their programming skills for Natural Language Processing applications |

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# Text books and Reference books

| | |
|---|---|
| T1 | Speech and Language processing: An introduction to Natural Language Processing, Computational Linguistics and speech Recognition by Daniel Jurafsky and James H. Martin |

| | |
|---|---|
| R1 | Manning and Schütze, Foundations of Statistical Natural Language Processing, MIT Press. Cambridge, MA |
| R2 | Neural Machine Translation by Philipp Koehn |
| R3 | Knowledge Graphs Methodology, Tools and Selected Use Cases by Dieter Fensel , Umutcan Şimşek, Kevin Angele, Elwin Huaman , Elias Kärle , Oleksandra Panasiuk , Ioan Toma, Jürgen Umbrich, and Alexander Wahler, Springer 2019 |
| R4 | Natural Language Toolkit. Bird and Loper, and other developers. Available for free at: – http://www.nltk.org/ |

# Evaluation Plan

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| Name | Weight |
|---|---|
| Quiz (best 2 out of 3) | 10% |
| Assignment 1 and 2 | 25% |
| Mid-term Exam | 30% |
| End Semester Exam | 35% |

**Please note that the Assignments (Experiential Learning) will also include situated learning aspect (related to area of work)**

Language is a massive, unstructured, global dataset powering real-world applications.
**Ubiquity of NL Data:** The "Uncharted Territory" Metaphor (The Vintage Map)
**Multilinguality & Low-Resource Languages**: 1000s of languages and dialects. Real-world applications require global coverage.
**Domain Diversity**: biomedical text vs. legal contracts vs. informal social media slang
**Bridge Core Tasks to Practical Applications Core Tasks**: Tokenization, POS Tag, NER, Parsing, Co- reference Resolution.
**NLP App**: End-to-end-> MT, IR, Summarization, Dialogue Systems /Agents, Sentiment /Market Analysis.

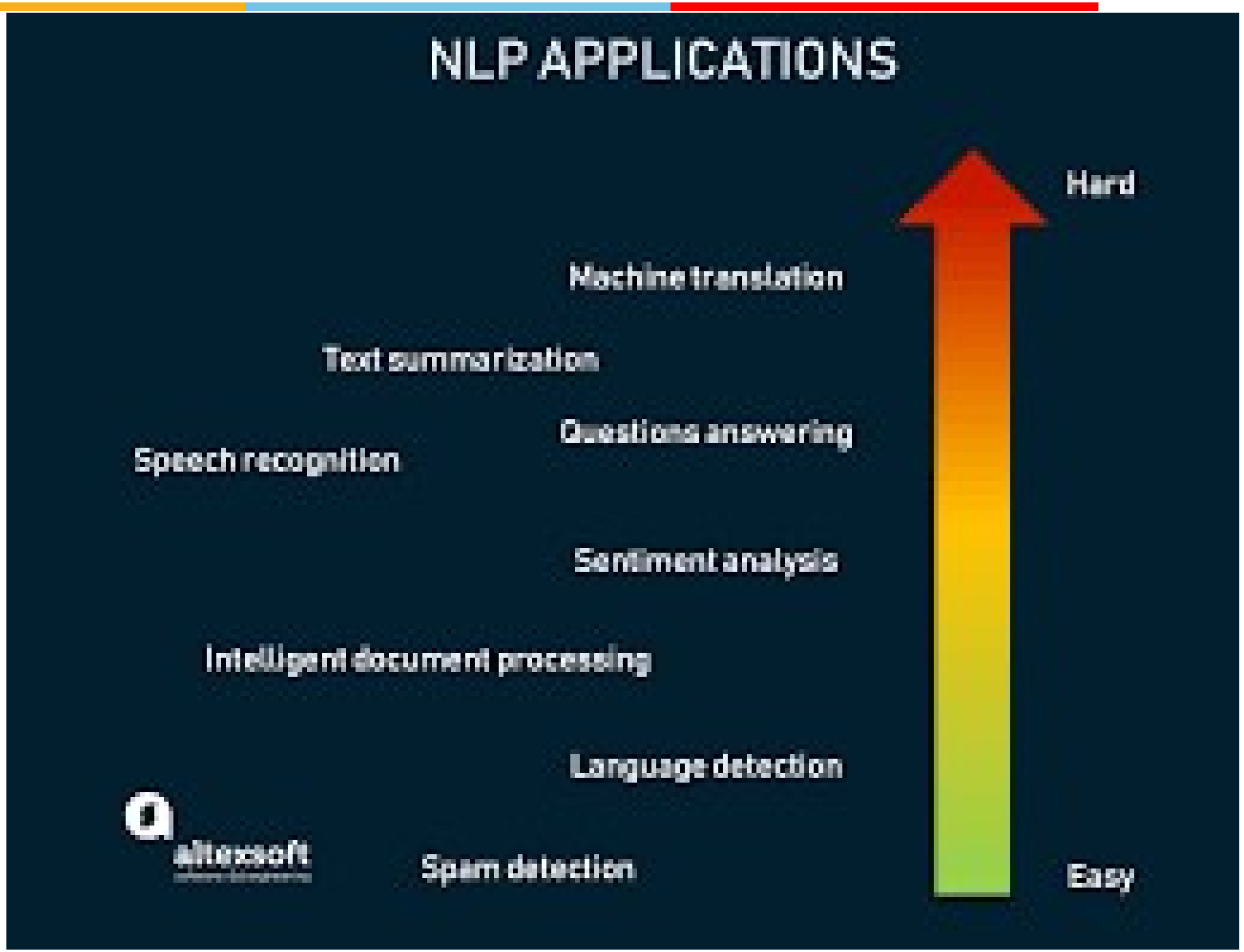
# NLP Tasks and Applications




It's a big world out there

And everyone uses language

# NLP Applications and Difficulty Level



# Application areas

- Text-to-Speech & Speech recognition
- Natural Language Dialogue Interfaces to Databases
- Information Extraction
- Document Image Analysis
- Automatic Summarization (https://pypi.org/project/sumy/)
- Text Proof-reading – Spelling & Grammar
- Machine Translation
- Fake News and Cyberbullying Detection
- Monitoring Social Media
- Plagiarism detection
- Question Answering System (https://haystack.deepset.ai/)
- Sentiment Analysis (https://aimultiple.com/open-source-sentiment-analysis)
- Amazon textract

NLP Applications span across domains like Healthcare, Finance, Manufacturing and many more

# NLP Applications by Industry

| Industry | Key applications | Use cases | Impact |
|---|---|---|---|
| Healthcare | Clinical NLP, EHR analysis, medical transcription | Extracting insights from physician notes, drug safety monitoring, diagnostic support | 87.5% reduction in documentation time; 95%+ accuracy in clinical records |
| BFSI | Fraud detection, sentiment analysis, compliance monitoring | Transaction pattern analysis, regulatory reporting, chatbots for customer service | 21.1% market share; 25% of institutions deployed NLP-based solutions by 2024 |
| E-Commerce | Product recommendations, chatbots, search optimization | Sentiment analysis, personalized marketing, returns processing | Improved customer engagement; double-digit loyalty metric boosts |
| Telecom | Customer service automation, intent detection | Call routing, complaint resolution, virtual assistants | Reduced support cost; faster query resolution |
| Manufacturing | Predictive maintenance, process documentation | Log analysis, fault detection, supply chain automation | Reduced downtime and operational costs |
| Legal / Enterprise | Contract analysis, knowledge management, ERP automation | Document summarization, compliance tracking, financial reporting | Faster document review; reduced manual workload |

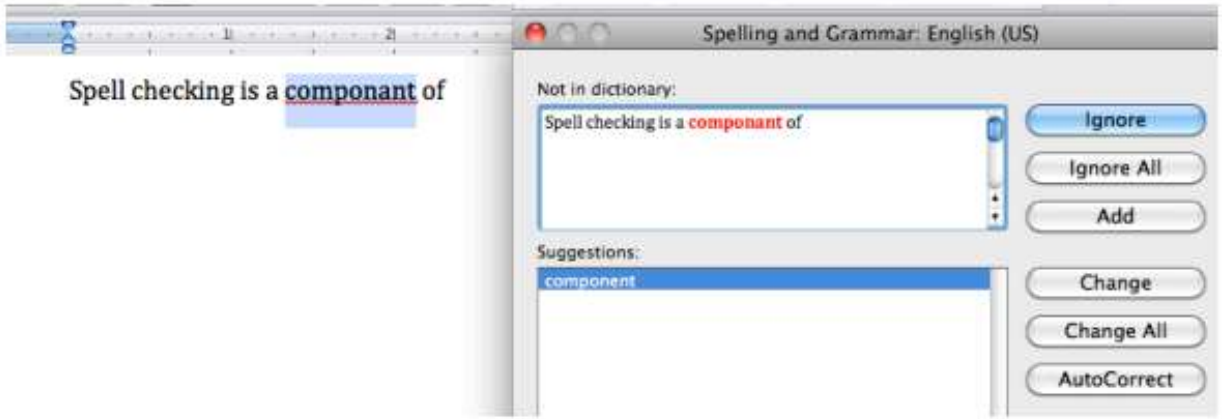
# NLP applications Growth

| Application | 5-yr growth. | Key skills required | Top job roles | Primary industries |
|---|---|---|---|---|
| **RAG applications** | 40–50%+ | •Vector DBs & semantic search<br>•LLM integration & prompt engineering<br>•Embedding models, LangChain<br>•Agentic / Graph RAG pipelines | RAG / GenAI Engineer, LLM Platform Engineer, AI Architect | Tech, Finance, Healthcare, Legal, Enterprise SaaS |
| **Sentiment analysis** | 25–35% | •Text classification, BERT fine-tuning<br>•Aspect & emotion-level analysis<br>•Real-time stream processing<br>•Multilingual model evaluation | NLP Engineer, Data Scientist (NLP), AI Analyst, ML Engineer | E-commerce, BFSI, Marketing, Healthcare |
| **Information extraction** | 25–35% | •NER, relation & event extraction<br>•Knowledge graph construction<br>•Schema-guided structured output<br>•Clinical / legal domain expertise | NLP / IE Engineer, Computational Linguist, AI Research Scientist | Healthcare, Legal, Defense, Finance, Government |
| **Machine translation** | 20–30% | •Seq2seq / multilingual transformers<br>•Domain-adaptive fine-tuning<br>•Low-resource language handling<br>•MT evaluation (BLEU, COMET) | MT Engineer, Localisation AI Engineer, NLP Research Scientist | Tech, Government, E-commerce, Media, Global Enterprises |
| **Grammar check** | 15–25% | •LLM fine-tuning & RLHF<br>•Error detection & style classification<br>•AI writing evaluation & red-teaming<br>•Prompt engineering | NLP Software Engineer, LLM Evaluator, AI Writing Tool Developer | EdTech, Publishing, SaaS, Enterprise Productivity |

# Spell check

## Applications for spelling correction

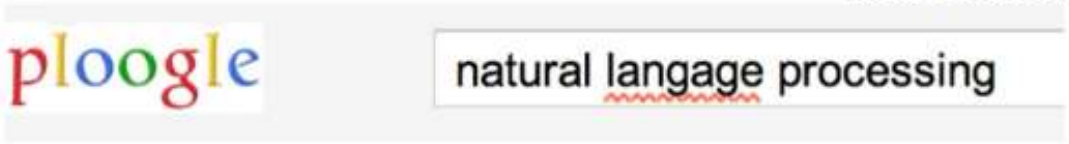
Word processing




Phones




Web search




2

Showing results for natural ***language*** processing
Search instead for natural langage processing

https://languagetool.org/

# Spell check and Grammar check

**Problem:**

... and he fired presidential **aid/aide** Dick Morris after ...

⇒ *aid* or

⇒ *aide*

**Training Data:**

| Spelling | Context |
|---|---|
| **(1) aid** | ... and cut the foreign *aid/aide* budget in fiscal 1996 ... |
| " " | ... they offered federal *aid/aide* for flood-ravaged states ... |
| **(2) aide** | ... fired presidential *aid/aide* Dick Morris after ... |
| " " | ... and said the chief *aid/aide* to Sen. Baker, Mr. John ... |

**Test Data:**

| Spelling | Context |
|---|---|
| ??? | ... said the longtime *aid/aide* to the Mayor of St. ... |
| ??? | ... will squander the *aid/aide* it receives from the ... |

# Case Study: AI-Powered Communication Enhancement




## Challenges

- Traditional tools often miss contextual errors (e.g., *your* vs. *you're*, *affect* vs. *effect*).
- Writers need more than just error correction; they need help communicating effectively and appropriately for their audience and context.

## SOLUTION

- **Hybrid Approach:** Combines rule-based checks, statistical NLP (analyzing patterns in vast text data), and advanced DL/Transformers/Agentic AI
- **Contextual Understanding:** AI models analyze the entire sentence and surrounding text to detect subtle errors and suggest contextually appropriate words.
- **Beyond Grammar:** Employs NLP to analyze tone (e.g., confident, formal, friendly), suggest improvements for clarity and conciseness, check for plagiarism, and offer full-sentence rewrites.
- **Continuous Learning:** Models are constantly updated based on user interactions and evolving language patterns

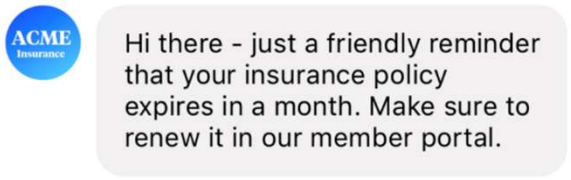
# Case Study: AI-powered Question Answering (QA) system



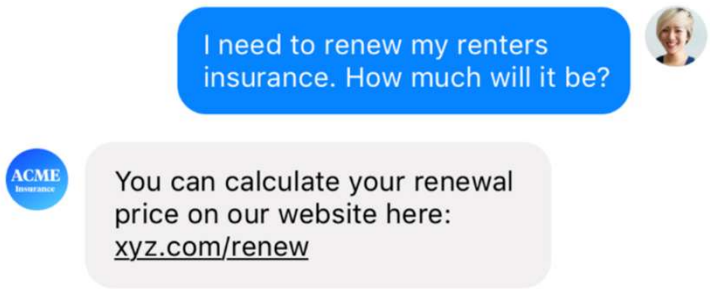
- 24/7 Demand, High volume, Long wait times
- **Training:** The AI was trained to accurately handle the top 100 most frequent customer questions.
- **Integration:** The QA chatbot was launched, acting as the *first point of contact* for all support inquiries.
- **Handoff):** A seamless "handoff" protocol was created, allowing the bot to escalate complex or sensitive issues to a live human agent, along with the full chat history.

# AI Assistants

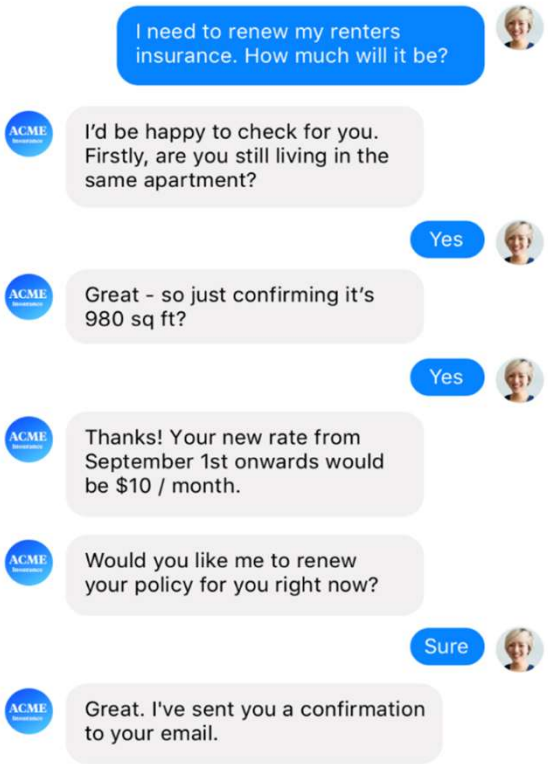
## Notification Assistant




## FAQ Assistant




## Contextual Assistant

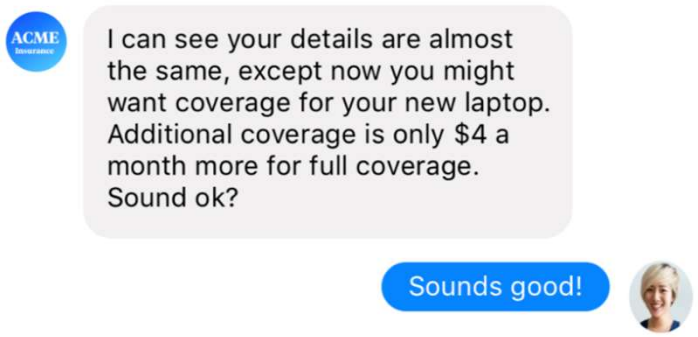


# AI Assistants

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## Personalized Assistant

- Assistant knows you much more in detail
- Quickly checks a few final things before giving you a quote tailored to your actual situation.




## Autonomous Organization of Assistants

- Group of AI assistants that know every customer personally
- Eventually run large parts of company operations—from lead generation over marketing, sales, HR, or finance



# Case Study: Automating Customer Support with NLP Chatbots

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- **Background**
- HDFC Bank, one of India's largest private sector banks, was facing increasing customer service demands. Traditional support channels (phone/email) were overburdened, resulting in delayed responses and reduced customer satisfaction.
- **Problem**
- High customer query volume (especially repetitive questions)
- Long response times
- High operational costs for customer support
- **Solution: EVA – HDFC Bank's AI Chatbot**

# Case Study: Automating Customer Support with NLP Chatbots



- **Key NLP Features Used**
  - **Intent recognition**: Classifies queries into categories (e.g., account balance, card issues)
  - **Named Entity Recognition (NER)**: Identifies dates, transaction amounts, account types
  - **Context management**: Maintains conversational state for multi-turn dialogues
  - **Multilingual support**: Handles queries in English and Hindi
- **Outcomes**
  - Handled over **2.7 million queries** in the first year
  - Resolved **85% of queries** without human intervention
  - **Reduced average query response time** to less than 0.4 seconds
  - Improved **customer satisfaction** and reduced cost-per-query
- **Challenges Faced**
  - Training the model on diverse and noisy real-world customer data
  - Maintaining user trust (transparency and security)
  - Escalating complex queries to human agents without losing context

# Retrieval-Augmented Generation

## Core RAG Mechanism

RAG integrates large language models with retrieval from external sources to improve response accuracy and reduce hallucinations.

## Typical RAG Pipeline Steps

The pipeline includes document ingestion, embedding creation, similarity search, and synthesizing responses.

## Enterprise QA Case Study

RAG enables secure, efficient querying of internal knowledge bases, enhancing employee productivity in enterprises.

## Agentic AI Orchestration

Agentic AI autonomously manages when to retrieve or generate content, optimizing user interactions based on intent.

# Case study – Knowledge Graph Applications in Agentic AI

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**Problem:** Large tech company, faced challenges with internal information access. Key data about projects, teams, expertise, and documentation was scattered across various internal wikis, project management tools (like Jira), employee directories, and shared drives.

Employees spent significant time searching for answers to complex questions like:

- "Who worked on the authentication module for Project X and what were the key technical documents?"
- "Which active projects are using Python and are related to our client Y?"
- "Find the design specification document for the UI redesign led by A's team."

# Enterprise Knowledge Graph (EKG) Construction



**Data Sources:** Internal Wiki, Jira, HR Database, Code Repositories (metadata), Document Management System.

**Entities:** Employee, Project, Team, Document, Skill, Client, CodeModule.

**Relationships:** WORKS_ON (Employee -> Project), MEMBER_OF (Employee -> Team), AUTHORED (Employee -> Document), RELATED_TO (Project -> Document), HAS_SKILL (Employee -> Skill), SERVES_CLIENT (Project -> Client), IMPLEMENTS (CodeModule -> Project).

**Population:** NLP techniques (NER, Relation Extraction) were used to extract entities and relationships from unstructured text (wikis, documents). Structured data was mapped directly (HR DB, Jira)

# Agentic AI System ("Ask Me")



**Core:** An LLM (like GPT-4 or a fine-tuned open-source model)

- **Tools:**
  - query_EKG: Takes a natural language query part and translates it into a structured query (e.g., Cypher for Neo4j) to fetch data from the EKG.
  - document_search: Performs keyword/semantic search over the document system for full-text retrieval when needed.
- **Capabilities:** Planning, Tool Use, Reasoning, Synthesis.

# Example of Enterprise KG application

**User Query:** "Who led the backend development for Project X and what recent documents did they author?"

**Agent's Plan:**

1.Identify entities: "Project X" (Project), "backend development" (implicit Skill/Role), "documents" (Document).

2.Find the team/person associated with "Project X" and "backend development lead" role via EKG. -> Use query_EKG.

3.Once the person is identified (e.g., “A"), find recent documents AUTHORED by “A" via EKG. -> Use query_EKG.

4.Synthesize the results into a natural language answer.

**Execution:**

•Agent calls query_EKG ("Find lead backend developer for Project X"). EKG returns "John Smith".

•Agent calls query_EKG ("Find recent documents authored by John Smith"). EKG returns a list: ["API_Design_v2.pdf", "Deployment_Strategy.docx"].

**Agent's Response:** “A led the backend development for Project X. Recent documents he authored include 'API_Design_v2.pdf' and Deployment_Strategy.docx'."

# Machine Translation

**English to Hindi** [http://anglahindi.iitk.ac.in](http://anglahindi.iitk.ac.in)

Simple Sentences.
*sarala vaakya .*

Welcome to London.
*landana men aapakaa svaagata hai*.

There are some cases which are still pending.

**NLP applications is an interesting course in MTech AIML of BITS WILP**

एनएलपी एप्लिकेशन बिट्स WILP के एमटेक एआईएमएल में एक दिलचस्प कोर्स है

# Machine Translation - Challenges

## Accent Restoration in Spanish & French

**Problem:**

**Input:** ... deja travaille cote a cote ...

⇓

**Output:** ... déjà travaillé côte à côte ...

**Examples:**

... appeler l'autre **cote** de l'atlantique ...
⇒ *côté* (meaning side) or
⇒ *côte* (meaning coast)

... une famille des **pecheurs** ...
⇒ *pêcheurs* (meaning fishermen) or
⇒ *pécheurs* (meaning sinners)

# Machine Translation - Challenges

## Capitalization

**Problem:**

... FRIED CHICKEN, **TURKEY** SANDWICHES AND FROZEN ...

⇒ *turkey* (the *bird*) or

⇒ *Turkey* (the *country*)

**Training Data:**

| Capitalization | Context |
|---|---|
| **(1) turkey** | ... OF FRIED CHICKEN , TURKEY SANDWICHES AND FROZEN ... |
| " " | ... NTS A POUND , WHILE TURKEY PRICES ROSE 1.2 CENTS ... |
| " " | ... PLAY , REAL GRADE-A TURKEY , WHICH ONLY A PRICE ... |
| **(2) Turkey** | ... INUNDATED EASTERN TURKEY AFTER THE EARLIER ... |
| " " | ... FEELINGS TOWARD TURKEY SURFACED WHEN GREECE ... |
| " " | ... THE CONTRACT WITH TURKEY WILL PROVIDE OPPORTU... |

**Test Data:**

| Capitalization | Context |
|---|---|
| ??? | ... NECK LIKE THAT OF A TURKEY ON A CHOPPING BLOCK ... |
| ??? | ... PROBLEM IS THAT TURKEY IS NOT A EUROPEAN ... |

# Machine Translation - Challenges

## Word sense disambiguation

**Problem:**

... He wrote the last **sentence** two years later ...

⇒ *sentencia* (legal sentence) or

⇒ *frase* (grammatical sentence)

**Training Data:**

| Translation | Context |
|---|---|
| **(1) sentencia** | ... for a maximum *sentence* for a young offender ... |
| " " | ... of the minimum *sentence* of seven years in jail ... |
| " " | ... were under the *sentence* of death at that time ... |
| **(2) frase** | ... read the second *sentence* because it is just as ... |
| " " | ... The next *sentence* is a very important ... |
| " " | ... It is the second *sentence* which I think is at ... |

**Test Data:**

| Translation | Context |
|---|---|
| ??? | ... cannot criticize a *sentence* handed down by ... |
| ??? | ... listen to this *sentence* uttered by a former ... |

# Machine Translation - Challenges

## Text to speech

**Problem:**

... slightly elevated *lead* levels ...

⇒ *lɛd* (as in *lead mine*) or

⇒ *li:d* (as in *lead role*)

**Training Data:**

| Pronunciation | Context |
|---|---|
| **(1) lɛd** | ... it monitors the *lead* levels in drinking ... |
| " " | ... conference on *lead* poisoning in ... |
| " " | ... strontium and *lead* isotope zonation ... |
| **(2) li:d** | ... maintained their *lead* Thursday over ... |
| " " | ... to Boston and *lead* singer for Purple ... |
| " " | ... Bush a 17-point *lead* in Texas , only 3 ... |

**Test Data:**

| Pronunciation | Context |
|---|---|
| ??? | ... median blood *lead* concentration was .. |
| ??? | ... his double-digit *lead* nationwide . The ... |

# Case Study: Scaling Global Communications

- **Custom Engines:** The system was trained on specific content to create custom models that understood its unique terminology and branding.

- **Centralized Platform:** All translation requests were funneled through the MT platform, which was integrated into existing content systems.

- **MT + Post-Editing:** The platform was used to instantly translate content. For high-visibility materials, human translators would then "post-edit" the AI's output, rather than translating from scratch.

# Information Extraction

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**As a task:** **Filling slots in a database from sub-segments of text.**

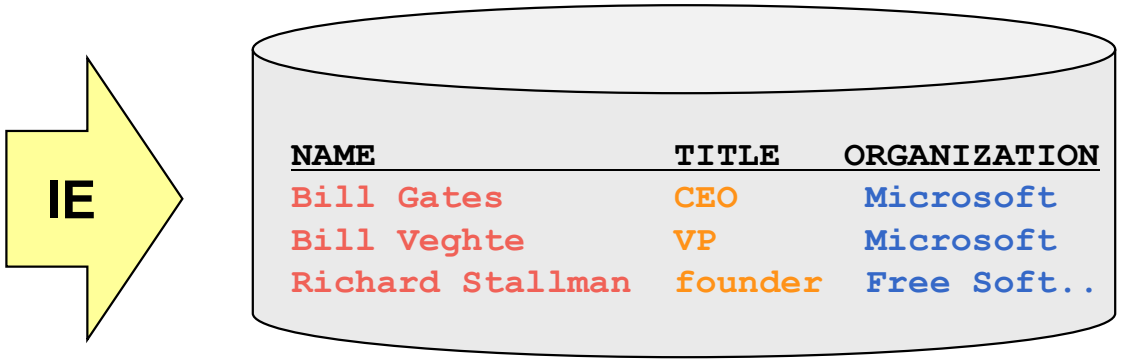
**October 14, 2002, 4:00 a.m. PT**

**For years, Microsoft Corporation CEO Bill Gates railed against the economic philosophy of open-source software with Orwellian fervor, denouncing its communal licensing as a "cancer" that stifled technological innovation.**

**Today, Microsoft claims to "love" the open-source concept, by which software code is made public to encourage improvement and development by outside programmers. Gates himself says Microsoft will gladly disclose its crown jewels--the coveted code behind the Windows operating system--to select customers.**

**"We can be open source. We love the concept of shared source," said Bill Veghte, a Microsoft VP. "That's a super-important shift for us in terms of code access."**

**Richard Stallman, founder of the Free Software Foundation, countered saying…**




Slide from Chris Brew, adapted from slide by William Cohen

# Phrase Types to Identify for IE

## Closed set

**U.S. states**

He was born in Alabama…

The big Wyoming sky…

## Regular set

**U.S. phone numbers**

Phone: (413) 545-1323

The CALD main office can be reached at 412-268-1299

## Complex pattern

**U.S. postal addresses**

University of Arkansas
P.O. Box 140
Hope, AR 71802

Headquarters:
1128 Main Street, 4th Floor
Cincinnati, Ohio 45210

## Ambiguous patterns, needing context and many sources of evidence

**Person names**

…was among the six houses sold by Hope Feldman that year.

Pawel Opalinski, Software Engineer at WhizBang Labs.

Slide from Chris Brew, adapted from slide by William Cohen

# Named Entity Recognition

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CHICAGO (AP) — Citing high fuel prices, United Airlines said Friday it has increased fares by $6 per round trip on flights to some cities also served by lower-cost carriers. American Airlines, a unit AMR, immediately matched the move, spokesman Tim Wagner said. United, a unit of UAL, said the increase took effect Thursday night and applies to most routes where it competes against discount carriers, such as Chicago to Dallas and Atlanta and Denver to San Francisco, Los Angeles and New York.

# NE Types

| Type | Tag | Sample Categories |
|---|---|---|
| People | PER | Individuals, fictional characters, small groups |
| Organization | ORG | Companies, agencies, political parties, religious groups, sports teams |
| Location | LOC | Physical extents, mountains, lakes, seas |
| Geo-Political Entity | GPE | Countries, states, provinces, counties |
| Facility | FAC | Bridges, buildings, airports |
| Vehicles | VEH | Planes, trains, and automobiles |

| Type | Example |
|---|---|
| People | *Turing* is often considered to be the father of modern computer science. |
| Organization | The *IPCC* said it is likely that future tropical cyclones will become more intense. |
| Location | The *Mt. Sanitas* loop hike begins at the base of *Sunshine Canyon*. |
| Geo-Political Entity | *Palo Alto* is looking at raising the fees for parking in the University Avenue district. |
| Facility | Drivers were advised to consider either the *Tappan Zee Bridge* or the *Lincoln Tunnel*. |
| Vehicles | The updated *Mini Cooper* retains its charm and agility. |

Slide from Jim Martin

# Example applications for IE

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- Classified ads
- Restaurant reviews
- Bibliographic citations
- Appointment emails
- Legal opinions
- Papers describing clinical medical studies
- …
- Adding facts to the semantic web

# Sentiment classification



What features of the text could help predict # of stars?
(e.g., using a log-linear model) How to identify more?
Are the features hard to compute? (syntax? sarcasm?)

★☆☆☆☆ **An extremely versatile machine!**, November 22, 2006

By **Dr. Nickolas E. Jorgensen "njorgens3"**

**This review is from: Cuisinart DGB-600BC Grind & Brew, Brushed Chrome (Kitchen)**

This coffee-maker does so much! It makes weak, watery coffee! It grinds beans if you want it to! It inexplicably floods the entire counter with half-brewed coffee when you aren't looking! Perhaps it could be used to irrigate crops... It is time-consuming to clean, but in fairness I should also point out that the stainless-steel thermal carafe is a durable item that has withstood being hurled onto the floor in rage several times. And if all these features weren't enough, it's pretty expensive too. If faced with the choice between having a car door repeatedly slamming into my genitalia and buying this coffee-maker, I'd unhesitatingly choose the Cuisinart! The coffee would be lousy, but at least I could still have children...

# Case Study: AI-powered sentiment analysis platform

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**Data Aggregation:**

•The tool pulled all reviews into a single dashboard.

**Aspect-Based Sentiment:**

•The AI didn't just label a review "Positive" or "Negative.“

• It identified specific topics (aspects) and the sentiment for each.

**Real-time Dashboard:**

- Operations managers could instantly filter feedback by property, region, date, and topic.

# Evaluation of NLP Applications

# Machine Translation Evaluation

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- **BLEU (Bilingual Evaluation Understudy):**
  - Compares n-gram overlaps between machine output and human reference translations.
  - Most widely used but penalizes creative phrasing.
- **METEOR:**
  - Considers synonyms, stemming, and paraphrases — more flexible than BLEU.
- **ROUGE:**
  - Recall-oriented; useful when coverage matters more than precision.
- **TER (Translation Edit Rate):**
  - Measures edits needed to convert MT output to a reference translation.
- **chrF:**
  - Character-level F-score; works well for morphologically rich languages.
- **COMET / BERTScore:**
  - Neural metrics using contextual embeddings; better correlation with human judgment.

# Spellcheck and Grammar check evaluation measure

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- **Precision:** Of all flagged errors, how many are real?
- **Recall:** Of all real errors, how many were caught?
- **$F_{0.5}$ Score:** Weighs precision higher than recall (used in CoNLL shared tasks).
- **ERRANT:** Framework for classifying and scoring grammatical error corrections by error type (spelling, tense, agreement, etc.).
- **MaxMatch ($M^2$) Scorer:** Standard for GEC (Grammatical Error Correction) benchmarks.

# IE evaluation measures

- **Precision, Recall, F1-Score:** Standard for entity/relation extraction tasks.
- **Exact Match vs. Partial Match:** Strict (full span must match) vs. lenient (overlap accepted).
- **MUC Scores (MUC-5/6/7):** Specifically for co-reference resolution and slot filling.
- **Entity-level F1:** Evaluates complete entity spans, not individual tokens.

# Sentiment Analysis evaluation measures

- **Accuracy:** Overall correct classifications.
- **Macro F1:** Averages F1 across all classes — important for imbalanced datasets.
- **Weighted F1:** Accounts for class frequency.
- **Confusion Matrix:** Reveals which sentiments are misclassified.

# Security, Privacy, Ethics, and Scalable NLP Systems




**Security and Privacy Concerns**

NLP systems must address risks like prompt injection, data leakage, and model misuse to protect user data and maintain trust.

**Ethics and Fairness**

Bias mitigation, fairness evaluation, and regulatory compliance are crucial for ethical and responsible AI adoption in NLP.

**Scalable NLP Architectures**

Efficient model architectures, distillation, quantization, and robust serving ensure low latency and cost-effectiveness at scale.

**Performance Monitoring and Evaluation**

Continuous monitoring, logging, and system evaluation maintain performance and trustworthiness in deployed NLP systems.

# Multimodal interfaces or modeling

1. Sign languages
2. Speech + gestures
3. Images + captions
4. Brain recordings, human reaction times

# The NLP Research Community

innovate achieve lead

- Software
  - Lots of people distribute code for these tasks
    - Search github – fun to download packages and play around!
    - Or you can email a paper's authors to ask for their code
    - PapersWithCode.com
    - Search for "awesome NLP" for some lists
  - Toolkits and end-to-end pipelines for text analysis
    - Hugging Face – > 1,143,998 models, > 248,002 datasets
      - Large pretrained models: `pip install transformers` (quick tour)
      - Task-specific models: `pip install allennlp`, etc.
    - Allen NLP (Python), Spacy (Cython), UDPipe (C++), Stanza (Python), CoreNLP (Java), NLTK (Python)



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# Thank you!!

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